HS-401: Professional Values and Ethics

LECTURE NO 6: *The Ethical Cycle*

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Case Gilbane Gold

1.1 Introduction (2/4)

The city of Gilbane has been processing its waste water into manure for agriculture for 75 years. This yields a tax benefit of $300 per year per household. Given this tax advantage, the manure produced is called "Gilbane Gold." For the past 15 years, the city also has a company producing computer parts: Z-Corp. The city attracted the company to come by offering tax benefits. The company is important for the city because it creates opportunities for employment. However, the production process produces lead and arsenic, which are emitted through the waste water of the factory.

Lead and arsenic are heavy metals that built up in organisms and may cause negative health effects. If concentrations of arsenic and lead would cumulate in Gilbane Gold, this might have negative long-term effects. Therefore, the restrictions that the city has set on the concentration of arsenic and lead in the waste water are about ten times as strict as Federal Regulations.

Independent environmental consultant Tom Richards, who has been hired by Z-Corp, has found out that the conventional method for measuring arsenic and lead in the waste water used by Z-Corp measures lower concentrations than a new, more reliable, method. However, the old method is the one prescribed by City Regulations and city officials, after being informed on the issue, have not objected to the continued use of that method. Moreover, Z-Corp can easily stay within the limits of the

1.1 Introduction (3/4)

City Regulations even with the new measurement method by diluting the waste water because the regulations only refer to concentrations and not to absolute amounts. Some consider this, however, "a major loophole in the law." When Richards further pursues the matter, Z-Corp decides not to renew his contract.

A young engineer, David Jackson, now becomes responsible for Z-Corp's emissions of arsenic and lead into the waste water. In the meantime, Z-Corp signs a contract with a Japanese firm which will result in a 500 percent increase in production. Jackson, who is really concerned now, raises the issue with management but is told that there is no money available to solve the problem: the factory is hardly profitable. Moreover, manager Diane Collins argues, as long as Z-Corp meets the law, it has no broader responsibility. Jackson nevertheless fears that the waste water treatment plant may not be able to deal with the larger amounts of arsenic and lead. Since he has a duty as professional engineer "to hold paramount the safety, health and welfare of the public," it might be appropriate to speak out in public. Indeed Jackson is approached by Channel 13 a local television station about the issue.

1.1 Introduction (4/4)

- Gilbane Gold is a *fictional* case in which a young engineer has to decide how to act in a difficult situation. It is the kind of situation you may also find yourself in after starting working as an engineer. Such situations call for moral judgment.
- However, moral judgment is not a straightforward process in which you simply apply ethical theories to find out what to do. Instead it is a process in which the formulation of the moral problem, the formulation of possible "solutions," and the ethical judging of these solutions go hand in hand.
- In this lecture we describe a systematic approach to problem-solving that does justice to the complex nature of moral problems and moral judgment: the ethical cycle.

1.2 Ill-Structured Problems

- ***Ill-structured problem***
  - Moral problem-solving is a messy and complex process, like a design process. In cases of ill-structured design problems, thinking about possible solutions will further clarify the problem and possibly lead to reformulation of the problem. Moreover, ill-structured problems may have several alternative (good, satisfying) solutions, which are not easily compared with each other. This is due to the fact that for ill-structured problems, no single criterion exists to order uniformly the possible solutions from best to worst.
  - The most important lesson to be learned from designing is that practical problem-solving is not only about analyzing the problem and choosing and defending a certain solution, but also about finding (new) solutions called as *synthetic reasoning*.

1.3 The Ethical Cycle (1/13)

- ***Ethical cycle-*** *A tool in structuring and improving moral decisions by making a systematic and thorough analysis of the moral problem, which helps to come to a moral judgment and to justify the final decision in moral terms.*
- Ultimately, moral problem-solving is directed at finding the morally best, or at least a morally acceptable, solution in a given situation in which a moral problem arises. It is, however, hard to guarantee that the ethical cycle indeed delivers such a solution, because people may reasonably disagree about what is the morally best, or a morally acceptable, solution.
- The ethical cycle consists of a number of "steps" (Figure 1 next slide). It is important to stress that by distinguishing these steps we do not want to suggest that moral problem-solving is a linear process. Rather, it is an iterative process, as the feedback loops in Figure 1 already suggests.

1.3 The Ethical Cycle (2/13)

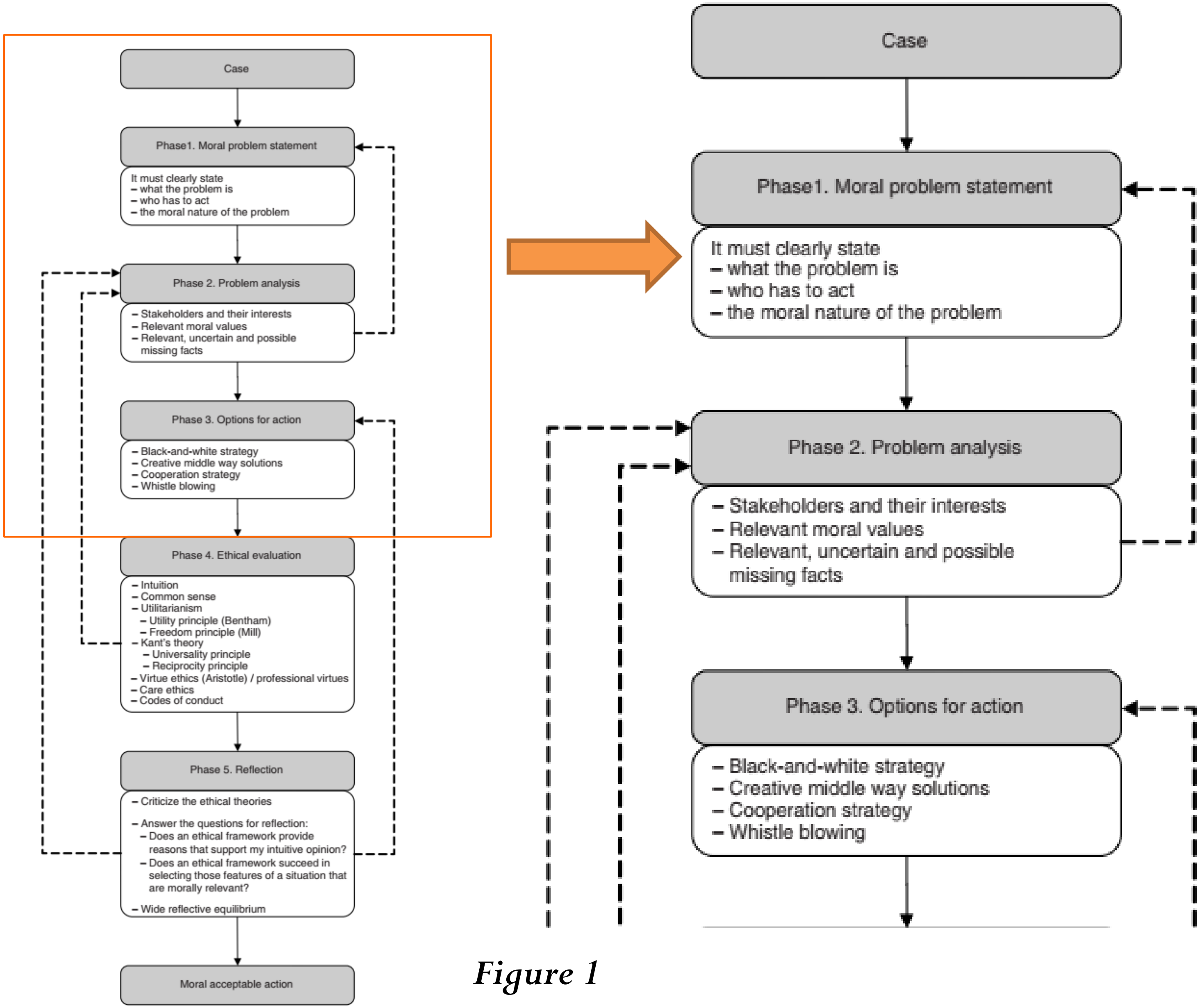

Figure 1

1.3 The Ethical Cycle (3/13)

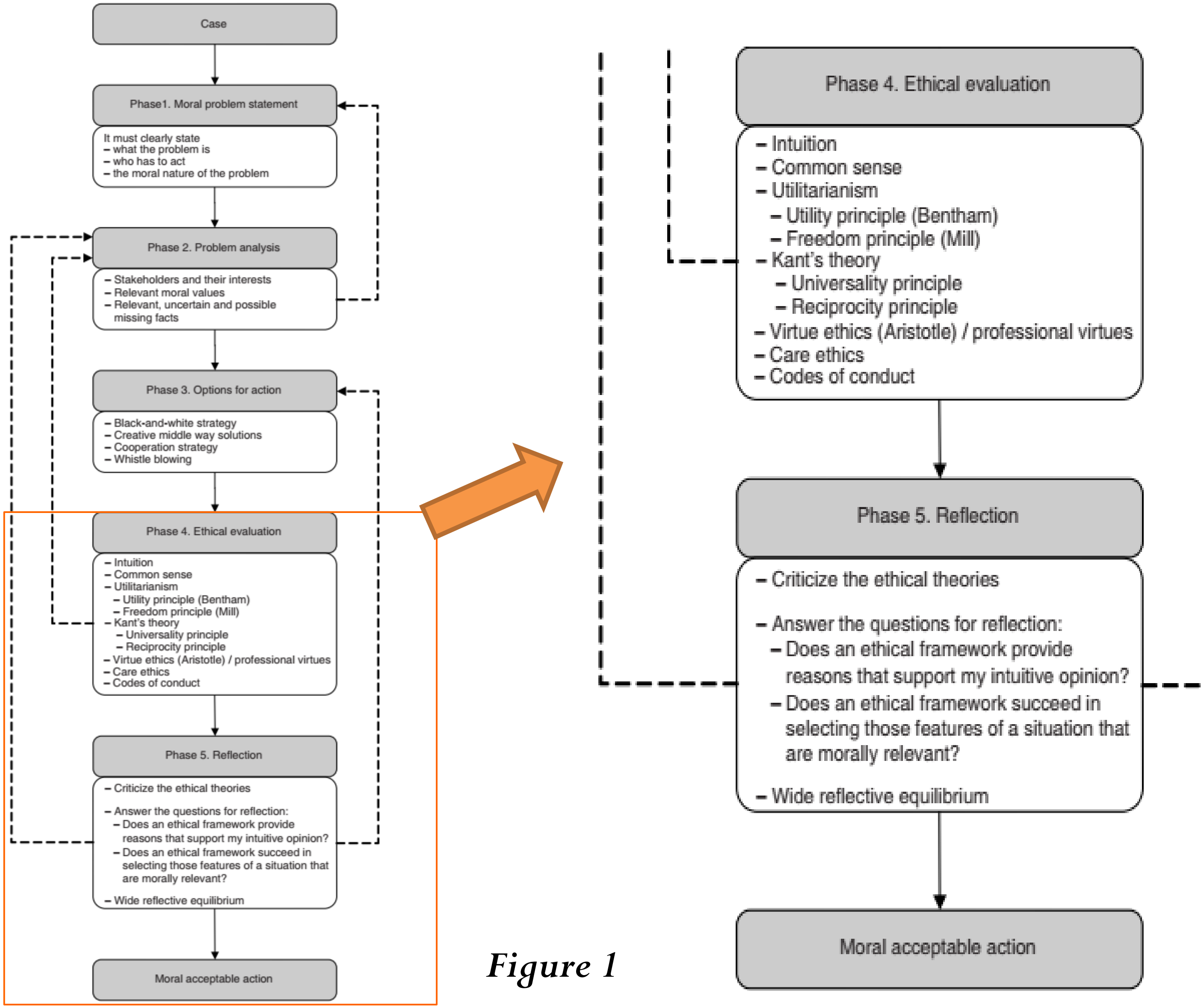

Figure 1

1.3 The Ethical Cycle (4/13)

- ***1.3.1 Moral problem statement***
  - The start of the ethical cycle is the formulation of a moral problem.

Gilbane Gold: Moral Problem Formulation

One possible problem formulation is:

- *Will the lead and arsenic in Z-Corp's waste water cause negative health effects?*

Another possible formulation is:

- *Should David inform the public about the potentially excessive levels of arsenic and lead in Z-Corp's waste water even if Z-Corp management does not consider it a serious problem?*

1.3 The Ethical Cycle (5/13)

- *1.3.2 Problem analysis*
  - During the problem analysis step, the relevant elements of the moral problem are described. Three important elements can be distinguished: *the stakeholders and their interests*, *the moral values that are relevant in the situation*, *and the relevant facts*. These elements are to be described during this step because they give a good impression of the current situation with respect to the moral problem; moreover, they are indispensable (absolutely necessary) for the carrying out of the later steps of the ethical cycle.

1.3 The Ethical Cycle (6/13)

Gilbane Gold: Relevant Values

- Public health
- Environmental care
- Public welfare
- Honesty (speaking the truth)
- Loyalty to the company
- Integrity (i.e., living by one's own moral standards and commitments)

These values can also be used to reformulate the moral problem a bit, for example:

> What should David Jackson do given on the one hand moral considerations of public health, environmental care, honesty, and integrity and on the other hand his loyalty to the company and the importance of Z-Corp for public welfare in Gilbane?

1.3 The Ethical Cycle (7/13)

Gilbane Gold: Main Stakeholders and their Interests

- *Management of Z-Corp*: increasing production in a profitable way, meeting legal requirements, good reputation
- *City officials and council*: protecting the safety of the inhabitants, maintaining income from Gilbane Gold, maintaining employment opportunities, protecting the environment
- *Farmers*: safe and reliable fertilizer
- *Inhabitants of city*: health, employment, low taxes, environmental protection
- *David Jackson:* being a reliable and honest engineer, keeping his job, meeting the law

1.3 The Ethical Cycle (8/13)

Gilbane Gold: Some Unknown or Disputed Facts

- The city waste water treatment plant will not be able to handle the increased levels of arsenic and lead in Z-Corp's waste water and this will cause environmental and health risks (contaminated vegetables).
- Expanding the waste water treatment of Z-Corp is too expensive. It would threaten the profits and might mean job losses or even bankruptcy.
- Jackson will lose his job if he goes public.

1.3 The Ethical Cycle (9/13)

- *1.3.3 Options for actions*
  - Often a moral problem is formulated in terms whether it is acceptable to engage in a certain action or not. In this ***black-and-white-strategy*** only two options for actions are considered, doing the action or not, other actions are simply not considered.
  - ***Cooperation Strategy*** The action strategy that is directed at finding alternatives that can help to solve a moral problem by consulting other stakeholders.
  - ***Whistle-blowing*** (speaking to the media or the public on an undesirable situation against the desire of the employer) is a last resort strategy because it usually brings large costs both to the individual employee and to the organization.

1.3 The Ethical Cycle (10/13)

Gilbane Gold: Options for Action

Our original problem formulation was:

> Should David inform the public about the potentially excessive levels of arsenic and lead in Z-Corp's waste water even if Z-Corp management does not consider it a serious problem?

This formulation suggests a black-and-white-strategy: either Jackson should inform the public or he should not. In this black-and-white strategy one of the options is whistle-blowing because it is clear that Z-Corp is against making the information public.

Now consider the reformulated problem:

> What should David Jackson do given on the one hand moral considerations of public health, environmental care, honesty, and integrity and on the other hand his loyalty to the company and the importance of Z-Corp for public welfare in Gilbane?

1.3 The Ethical Cycle (11/13)

Gilbane Gold: Options for Action

This formulation suggests a range of other options, including:

1. Develop a better but inexpensive treatment method;
2. Contact the engineering society for advice and help;
3. Contact city council to inform them about the new problem and ask them to take action; or
4. Contact people from the city's waste water treatment plant to see how serious the problem is.

1.3 The Ethical Cycle (12/13)

- *1.3.4 Ethical evaluation*
  - Ethical evaluation also can be based on more informal ethical frameworks. We distinguish two such frameworks here: intuitions and common sense.
  - ***Intuitivist framework-*** The ethical framework in which options for action are evaluated on basis of one's view about what is intuitively most acceptable and that formulates arguments for this statement.
  - ***Common sense method-*** This method asks to weigh the available options for actions in the light of the relevant values.

1.3 The Ethical Cycle (13/13)

- *1.3.5 Reflection*
  - Since the different ethical frameworks, including the informal frameworks, do not necessarily lead to the same conclusion, a further reflection on the outcomes of the previous step is usually required. The goal of this reflection is to come to a well-argued choice among the various options for actions, using the outcomes of the earlier steps.
  - The approach to reflection we want to advocate here is known as the method of wide reflective equilibrium. This approach aims at making coherent three types of moral beliefs:
    1. Considered moral judgments;
    2. Moral principles; and
    3. Background theories

1.4 Collective Moral Deliberations and Social Arrangements (1/2)

- The emphasis in the ethical cycle is on individual judgment. However, in many, if not most, situations in real life, other people will be involved in and affected by your choices.
- It seems doubtful that every person using the ethical cycle would come to the same conclusion as you did.
- We therefore propose a more practical solution – engaging in a moral deliberation with other people involved and possibly affected. A certain support from others is therefore required to be able to act in a morally effective way.
- The final step in the ethical cycle is reflection, leading to a well-argued choice for an option of action. This choice, however, needs not be your final choice; it can also be seen as a provisional choice that you can revise in discussion with others.

1.4 Collective Moral Deliberations and Social Arrangements (2/2)

- So conceived, deliberation is mainly a tool to improve one's moral judgment.
- ***Moral deliberation-*** An extensive and careful consideration or discussion of moral arguments and reasons for and against certain actions.

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